

Dataset source

Project: Preparing customer data for churn classification

Dataset: Telco Customer Churn

Kaggle publisher: BlastChar

Original Kaggle URL: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Downloaded: September 20, 2026

Kaggle version: 1 (published February 23, 2018)

Original filename: WA_Fn-UseC_-Telco-Customer-Churn.csv

What the data represents

Each row represents one customer in an IBM sample dataset. The original CSV has 7,043 customer records and 21 columns, meeting the assignment's 2,000-row requirement. The target is churn: whether the customer left within the last month (Yes/No). This is an educational sample, not evidence about a current real-world customer population.

The Kaggle description identifies customer services, account information, and demographic attributes. It links to IBM's expanded sample documentation, which describes a fictional telecommunications company. This project uses the original 21-column Kaggle file, not the expanded IBM files or their additional churn scores and reasons.

Fields	Meaning
customerID	Customer identifier; retain for tracing rows, exclude from predictors
gender, SeniorCitizen, Partner, Dependents	Customer demographic and household attributes
tenure	Number of months as a customer
PhoneService, MultipleLines	Telephone service subscriptions
InternetService	Internet service type or no internet service
OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport	Internet add-on services
StreamingTV, StreamingMovies	Streaming subscriptions
Contract	Contract term
PaperlessBilling, PaymentMethod	Billing and payment arrangements
MonthlyCharges, TotalCharges	Monthly and cumulative charges
Churn	Classification outcome: Yes or No

SeniorCitizen is a binary indicator (0/1). Values such as No internet service and No phone service are meaningful categories, not missing values. The notebook checks numeric text conversion and missing entries using training data only.

Provenance and preservation

The supplied raw file was extracted directly from the Kaggle dataset download archive. It has not been manually edited. All split, cleaning, encoding, transformation, and balancing operations are recorded in 03_eda_preprocessing.ipynb.

- Download endpoint: <https://www.kaggle.com/api/v1/datasets/download/blastchar/telco-customer-churn>
- Metadata endpoint: <https://www.kaggle.com/api/v1/datasets/view/blastchar/telco-customer-churn>

- Raw file size: 977,501 bytes
- Raw file SHA-256: 88be4b93fbe0cc83421af1c503794c97c342eca914c1576db7c276e61d61358a
- Kaggle rights label: Data files © Original Authors. Original data attribution is retained; this project does not assign a new license to the source data.

The notebook exports a raw 30% holdout before exploring training data and verifies that both the original source and holdout remain unchanged. The final cleaned dataset is **the training partition only**.

References

1. BlastChar. Telco Customer Churn. Kaggle.
2. IBM Samples Team. Telco customer churn (11.1.3+).
3. scikit-learn. Common pitfalls: data leakage.

Instructor approval: Approved on September 27, 2026, as reported by Viviana.